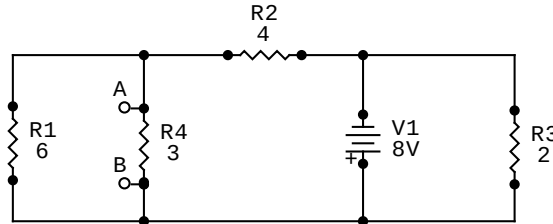
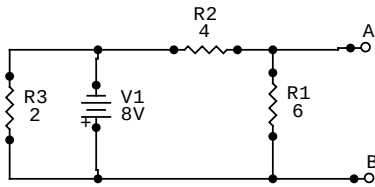


ENPH 334/PHYS 334 – Solutions to Class Problems, Week 1

1. Determine the Thevenin equivalent circuit between points A & B.

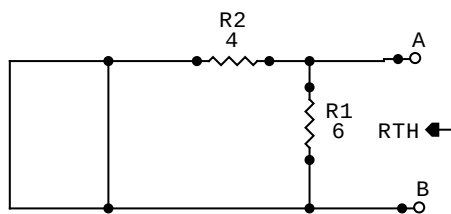


Re-draw the circuit with the source on the left → the “normal” way.



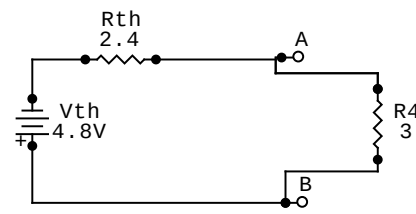
$$V_{TH} = \frac{R1}{R1 + R2} V1 = \frac{6}{6 + 4} (-8V) = -4.8V$$

R_{TH} :- Short $V1$ (and hence $R3$):-



$$R_{TH} = 4\Omega \parallel 6\Omega = 2.4\Omega$$

Hence, the Thevenin circuit:-



2. Required:- Voltage across $R2$ and Current through $R1$

First, short source $V2$:-

$$R2 \parallel R3 = 2\Omega$$

$$V_{R2} = (2/(6+2)) \times 5V = 1.25V$$

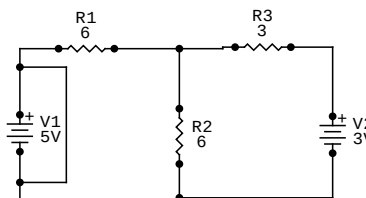
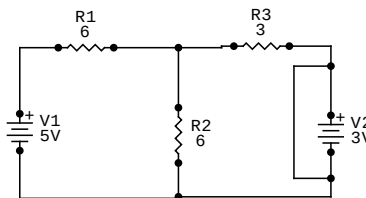
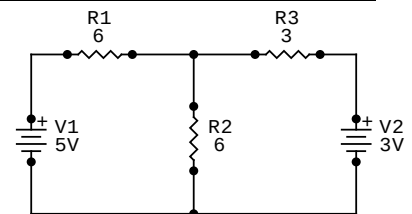
Next, short $V1$:-

$$R1 \parallel R2 = 3\Omega$$

$$V_{R2} = (3/(3+3)) \times 3V = 1.5V$$

Adding, $V_{R2} = 1.25 + 1.5 = \underline{2.75V}$

$$I_{R1} = (5 - V_{R2})/R1 = 2.25V/6\Omega = 0.375A \rightarrow \text{i.e. L to R}$$



3. **Series R-C circuit, with $R = 5 \text{ k}\Omega$ and $C = 100 \text{ nF}$. $V_S = 12 \text{ V}$ at 500 Hz .
The Phasor diagram is shown below:-**

The impedance of the series R - C Circuit, $Z = R - \frac{j}{\omega C}$, and $|Z| = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$

$$R = 5000 \Omega, \text{ and } \frac{1}{\omega C} = \frac{1}{2\pi(500)(100 \times 10^{-9})} = 3183 \Omega, \text{ so that } |Z| = 5927 \Omega$$

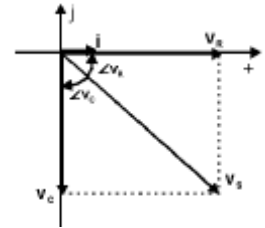
$$\therefore i = \frac{12 \text{ V}}{5927 \Omega} = 2.025 \text{ mA}$$

$$V_R = 2.025 \text{ mA} \times 5000 \Omega = 10.125 \text{ V}$$

$$V_C = 2.025 \text{ mA} \times 3183.5 \Omega = 6.446 \text{ V}$$

V_R is the angle between i & V_S , as seen from the phasor diagram

$$\angle V_R = \tan^{-1} \frac{V_C}{V_R} = \tan^{-1} \left(\frac{6.446}{10.125} \right) = +32.48^\circ$$

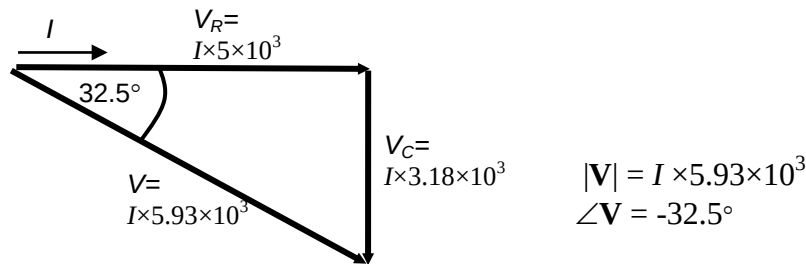


Alternative solution:-

6.24 Voltage across resistor $V_R = IR = I \times 5 \times 10^3$.

Reactance of capacitor $X_C = 1/\omega C = 1/(2\pi \times 500 \times 100 \times 10^{-9}) = 3.18 \text{ k}\Omega$.

Therefore voltage across capacitor $V_C = I X_C = I \times 3.18 \times 10^3$.



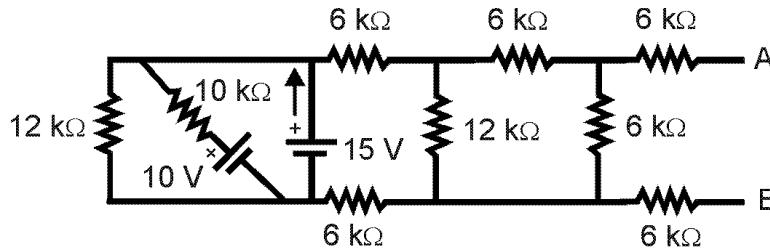
Since the voltage across the RC combination is known to be 12 V , it follows that

$$I \times 5.93 \times 10^3 = 12$$

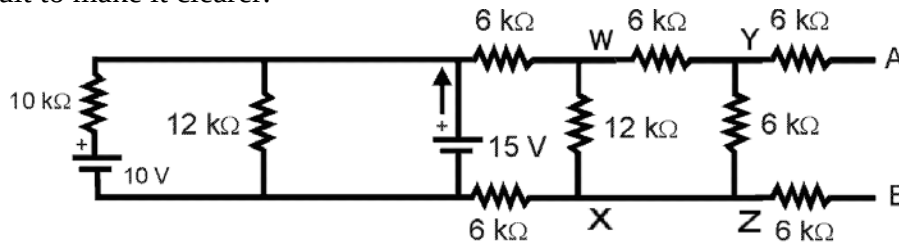
$$I = \frac{12}{5.93 \times 10^3} = 2 \text{ mA}$$

Therefore the magnitude of the current is 2 mA and voltage lags the current by 32.5° (or alternatively, the current leads the voltage by 32.5°).

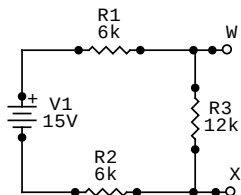
ENPH334/PHYS 334 -- Week 2 Thevenin Reduction



Redraw the circuit to make it clearer.



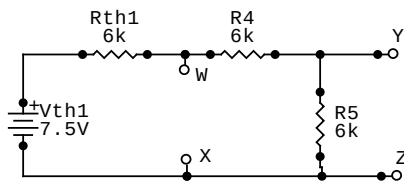
- Thevenin reduction at (A,B) -- the circuit to the left of the 15 V battery has no effect in this case.
- Use successive Thevenin reductions, using small circuit blocks



1) Consider points (W,X):-

$$V_{th1} = V_{WX} = [12k\Omega / (12+6+6)k\Omega] \times 15V = 7.5V$$

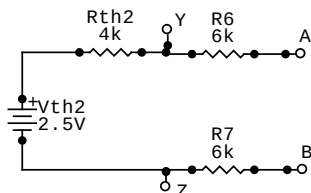
$$R_{th1} \text{ (short } V1) \rightarrow 12k\Omega \parallel (6+6)k\Omega = 6k\Omega$$



2) Consider points (Y,Z):-

$$V_{th2} = V_{YZ} = [6k\Omega / (6+6+6)k\Omega] \times 7.5V = 2.5V$$

$$R_{th2} \text{ (short } V_{th1}) \rightarrow 6k\Omega \parallel (6+6)k\Omega = 4k\Omega$$

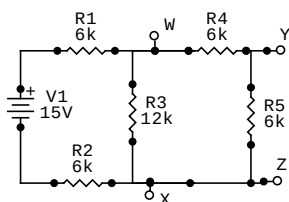


3) Consider points (A,B):-

$$V_{TH} = V_{AB} = V_{th2} = \underline{2.5V}$$

$$R_{TH} \text{ (short } V_{th2}) \rightarrow (6+4+6)k\Omega = \underline{16k\Omega}$$

Current from the 15 V battery:- The final circuit (3) is open-circuit. Therefore $I = 0$.



For circuit (2), $I_{R5} = 2.5V/6k\Omega = 0.4167mA = I_{R4}$, since $R4$ and $R5$ are in series.

And, in this case, $I_{R3} = 0.4167mA$ also (current divider rule)

Therefore, $I_{R1} = 0.4167mA + 0.4167mA = 0.8334mA$ (K.C.L.)

Now consider the circuit to the left of the 15 V battery:-

$$I_{12k\Omega} = 15V/12k\Omega = 1.25mA$$

and $I_{10k\Omega} = (15V-10V)/10k\Omega = 0.5mA$, since this resistor has 15V on one end, and 10V on the other end, i.e. 5V across it.

Therefore **total battery current** = $(0.8334 + 1.25 + 0.5) mA = \underline{2.584mA}$